

Exercise 1: Control Structures

Scenario 1: The bank wants to apply a discount to loan interest rates for customers above 60 years old.

- **Question:** Write a PL/SQL block that loops through all customers, checks their age, and if they are above 60, apply a 1% discount to their current loan interest rates.

Solution:- `DECLARE CURSOR c_customers IS SELECT CustomerID, Age FROM Customers;
BEGIN FOR cust IN c_customers LOOP
IF cust.Age > 60 THEN UPDATE Loans
SET InterestRate = InterestRate - 1
WHERE CustomerID = cust.CustomerID;
END IF;
END LOOP;`

Scenario 2: A customer can be promoted to VIP status based on their balance.

- **Question:** Write a PL/SQL block that iterates through all customers and sets a flag IsVIP to TRUE for those with a balance over \$10,000.

Solution:-`DECLARE CURSOR c_customers IS SELECT CustomerID, Balance FROM Customers;
BEGIN FOR cust IN c_customers LOOP
IF cust.Balance > 10000 THEN UPDATE Customers
SET IsVIP = 'TRUE' WHERE CustomerID = cust.CustomerID;
END IF;
END LOOP;`

Scenario 3: The bank wants to send reminders to customers whose loans are due within the next 30 days.

- **Question:** Write a PL/SQL block that fetches all loans due in the next 30 days and prints a reminder message for each customer.

Solution:- `DECLARE CURSOR c_loans IS SELECT CustomerID, LoanID, DueDate FROM Loans WHERE DueDate BETWEEN SYSDATE AND SYSDATE + 30;
BEGIN FOR loan IN c_loans LOOP
DBMS_OUTPUT.PUT_LINE('Reminder: Customer ' || loan.CustomerID || ' has Loan ' || loan.LoanID || ' due on ' || TO_CHAR(loan.DueDate, 'DD-MON-YYYY'));
END LOOP;
END;`